

ATLAS: Anytime-Valid Certification of World-Model Faithfulness via E-Process Betting

Highlights

- ATLAS gives world models a running, anytime-valid faithfulness certificate.
- Per-horizon betting e-processes need no exchangeability or i.i.d. data.
- The faithfulness frontier yields a trust horizon that clips planning depth.
- Proves uniform validity, an anytime simulation lemma, and delay optimality.
- Real driving, drone, and neural world-model tests confirm the theory holds.